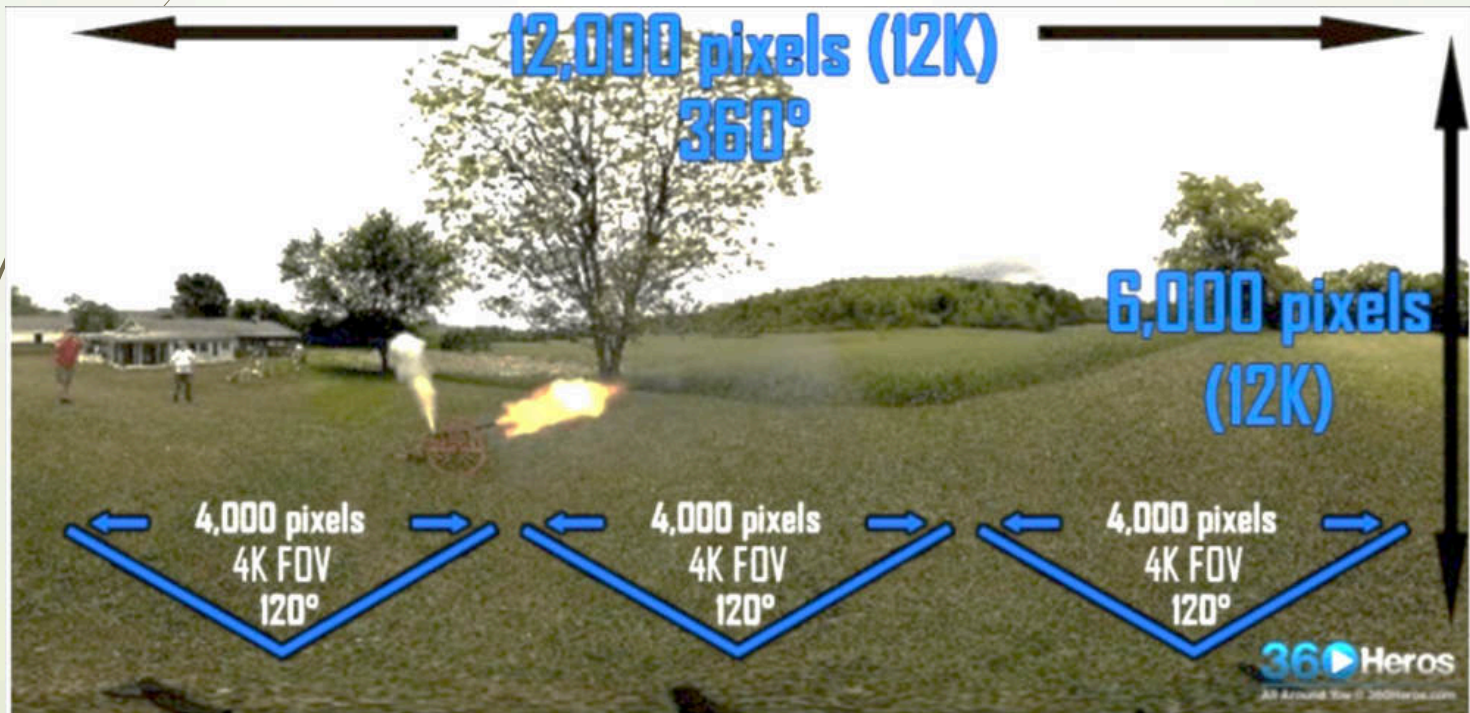


Agenda

- Cameras and Displays
- Projection Models
- Tiled Streaming (PA1)
- Edge Rendering
- Fixation Prediction
- Other Challenges

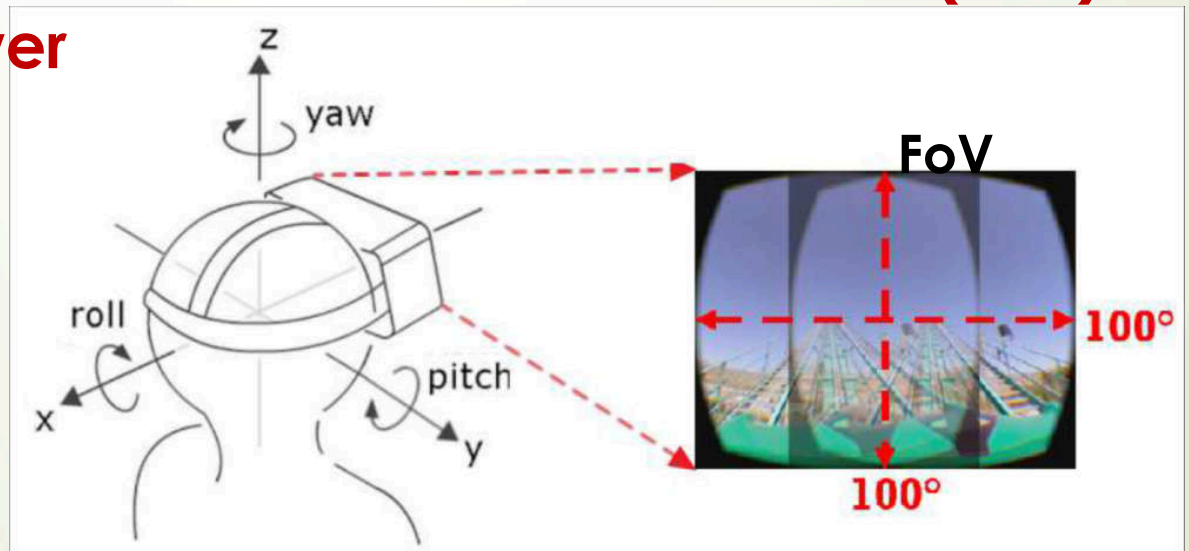
Streaming 360 Videos is Difficult, Because...

- 4k resolution in HMD requires 12k resolution for the whole 360° video (≈ 135 Mbps in HEVC )
- extremely large file size ⇒ **insufficient bandwidth**



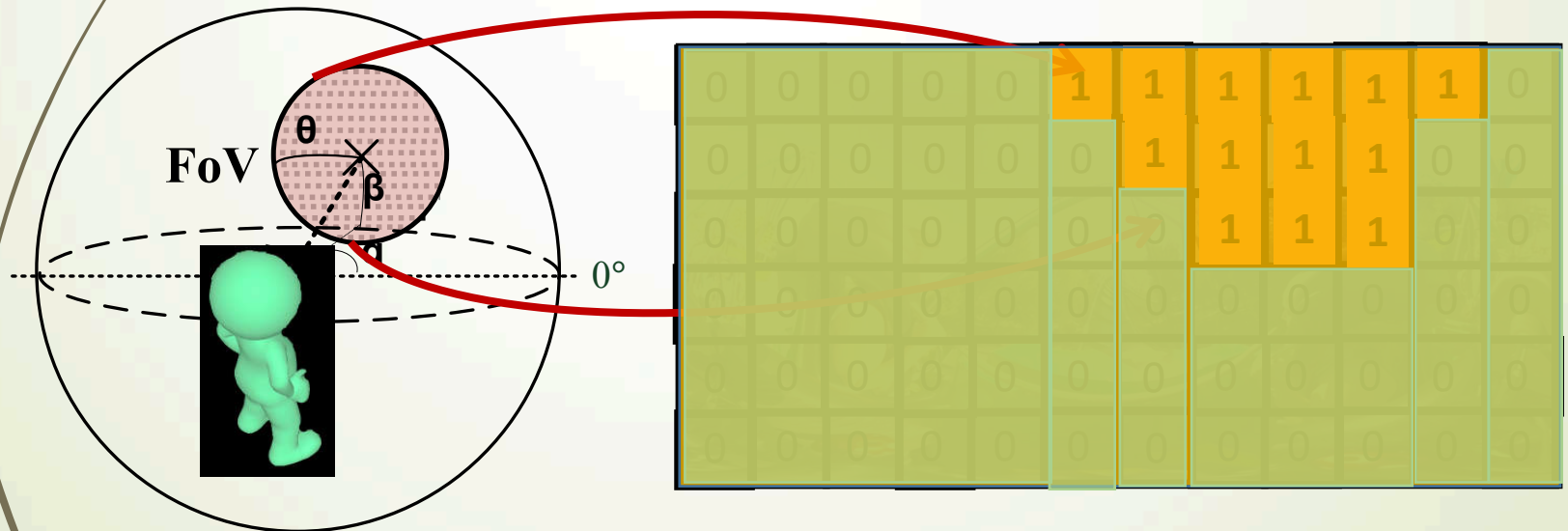
Potential Solution

- The HMD viewer only gets to see a small part of the whole 360° video ($< 1/3$)
 - The viewer actively changes the viewing orientation when rotating his/her head.
- ⇒ **Only stream the current Field-of-View (FoV) of the viewer**



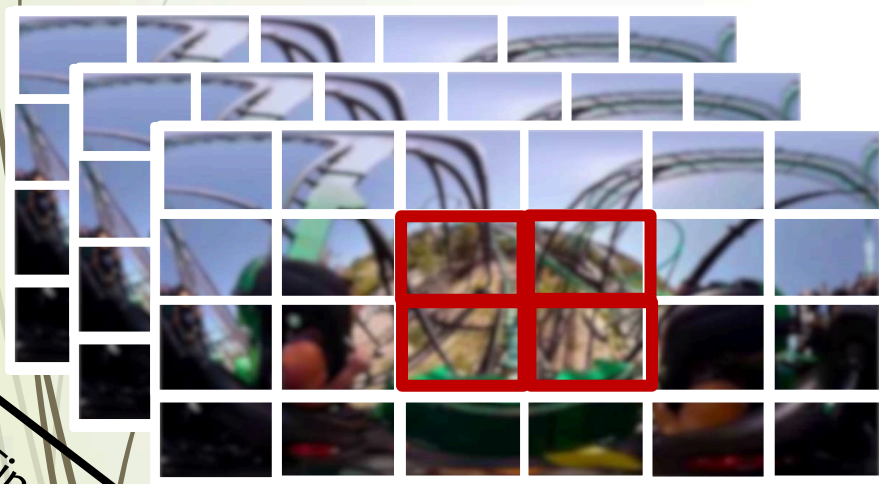
Tiled Encoding and (DASH) Streaming

- 360° video is split into tiles of sub-videos (spatial) and independently encoded
 - Only the tiles overlapped with the viewer's FoV are streamed to the client

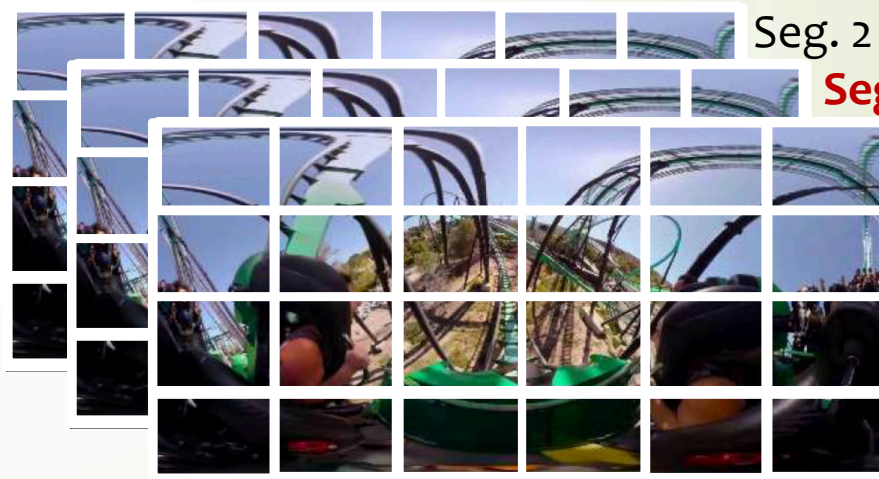


Tiled Segments in Different Quality

- Tiles are split into temporal segments
- tile qualities can change in every segments



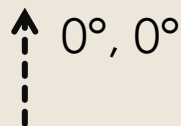
Low-quality



High-quality

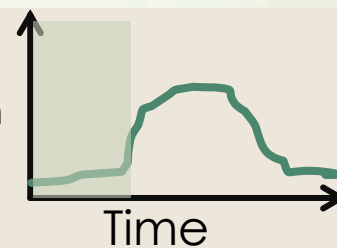


Selected tiles

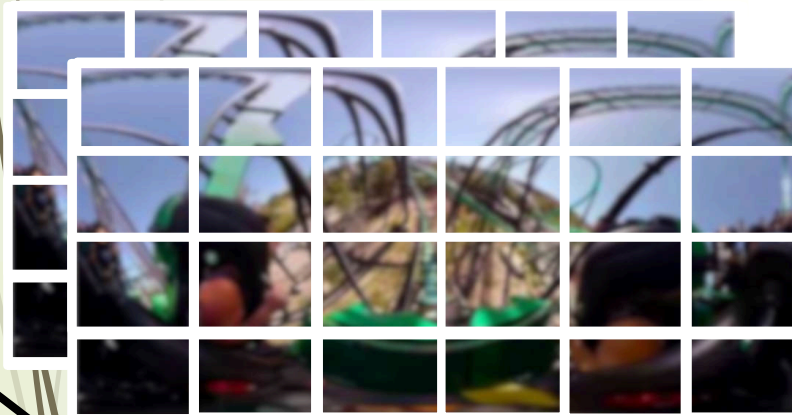


Viewer Orientation

Available
Bandwidth



Adaptive to Bandwidth and Viewer Orientation



Low-quality



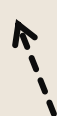
Seg. 3

Seg. 2

High-quality



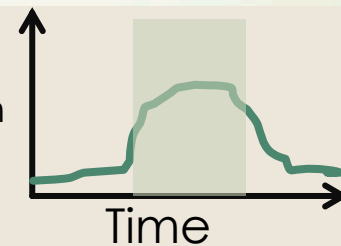
Selected tiles



$-15^\circ, 75^\circ$

Viewer Orientation

Available
Bandwidth



Time